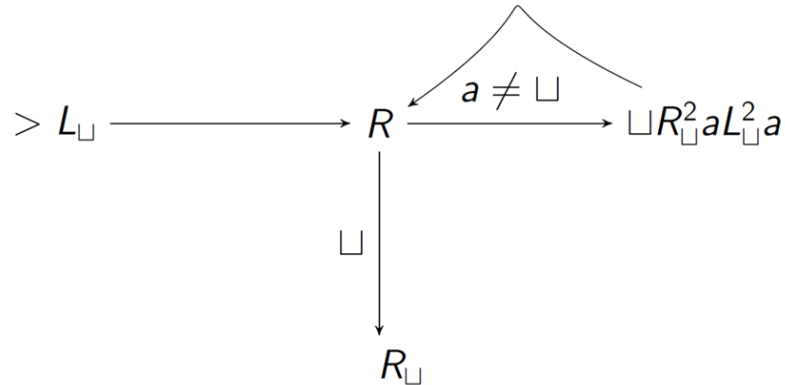


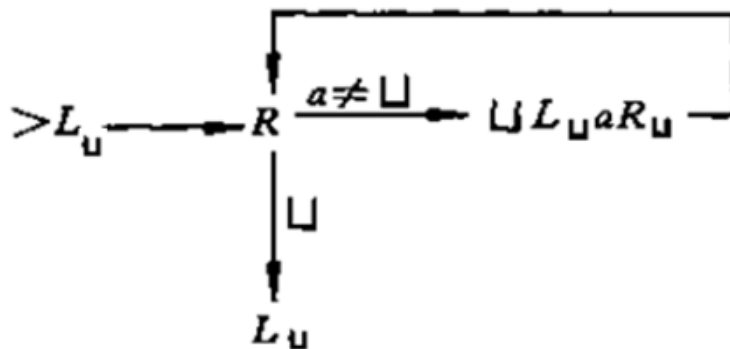
Lab 2-1: Turing machine computation

1. Based on the following copy Turing machine, give the sequence of tape changes on input $\sqcup aa \sqcup$.



Note: This Turing machine is given on page 52 of the PPT for Chapter 2.

2. Based on the following left-shifting machine, give the sequence of tape changes on input $\sqcup aa \sqcup$.



Note: This Turing machine is given on page 53 of the PPT for Chapter 2.

3. Answer the following questions given in the book “ Introduction to the Theory of Computation by Michael Sipser ” . (Pages 159-161)

Note: Although the solutions are given in the book, you are required to give the detailed solution by yourself. Copying the solution in the book will lead to zero marks.

3.1 a.-c.

Note: using the configuration format given in the class, such as $(q_1, w1a1v1)$.

3.1 This exercise concerns TM M_2 whose description and state diagram appear in Example 3.7. In each of the parts, give the sequence of configurations that M_2 enters when started on the indicated input string.

- a. 0.
- ^Ab. 00.
- c. 000.
- d. 000000.

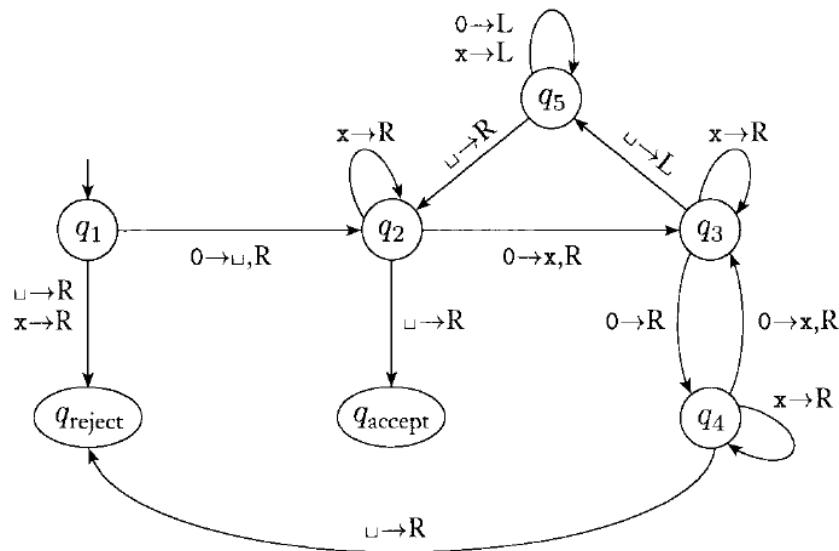


FIGURE 3.8

State diagram for Turing machine M_2

In this state diagram, the label $0 \rightarrow \square, R$ appears on the transition from q_1 to q_2 . This label signifies that, when in state q_1 with the head reading 0, the machine goes to state q_2 , writes $\square$, and moves the head to the right. In other words, $\delta(q_1, 0) = (q_2, \square, R)$. For clarity we use the shorthand $0 \rightarrow R$ in the transition from q_3 to q_4 , to mean that the machine moves to the right when reading 0 in state q_3 but doesn't alter the tape, so $\delta(q_3, 0) = (q_4, 0, R)$.